

Word2Vec

EHSG

✓ **Definition:**

- ✓ Word2vec is a technique in natural language processing for obtaining vector representations of words.
- ✓ These vectors capture information about the meaning of the word based on the surrounding words.
- ✓ The word2vec algorithm estimates these representations by modeling text in a large corpus.
- ✓ Once trained, such a model can detect synonymous words or suggest additional words for a partial sentence.
- ✓ Word2vec was developed by [Tomáš Mikolov](#), Kai Chen, Greg Corrado, [Ilya Sutskever](#) and Jeff Dean at Google, and published in 2013.

- ✓ Word2vec represents a word as a high-dimension vector of numbers which capture relationships between words.
- ✓ In particular, words which appear in similar contexts are mapped to vectors which are nearby as measured by cosine similarity.
- ✓ This indicates the level of semantic similarity between the words;
- ✓ so for example the vectors for *walk* and *ran* are nearby, as are those for "but" and "however", and "Berlin" and "Germany".

- CBOW
- SKIP-Gram

CBOW

we will find the center with near by words.

- ## Continuous Bag of Words

The product is really good

The product is wonderful

The product is awful

Continuous Bag of Words

The product is really good The product is wonderful The product is awful

Concatenating the all the sentences, to make a one single senense.

The product is really good The product is wonderful The product is awful

↑
Target

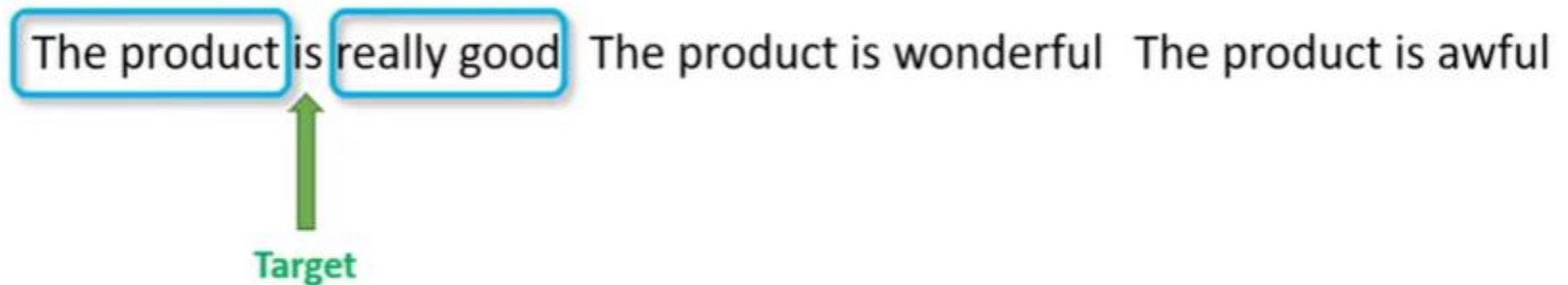
Window Size

1

Choose Window size, Here (The, is) = input words & Product as Target word.

Continuous Bag of Words

The product is really good The product is wonderful The product is awful



Target

Window Size

2

Step - 1

The product is really good The product is wonderful The product is awful

Step - 2

Window Size 1

Step - 3

Get one hot encoding for each word

Step - 1

The product is really good The product is wonderful The product is awful

Step - 3

1	The							
2	product							
3	is							
4	really							
5	wonderful							
6	good							
7	awful							

Step - 1

The product is really good The product is wonderful The product is awful

Step - 3

1	The	1						
2	product		1					
3	is			1				
4	really				1			
5	wonderful					1		
6	good						1	
7	awful							1

Step - 1

The product is really good The product is wonderful The product is awful

Step - 3

1	The	1	0	0	0	0	0	0
2	product	0	1	0	0	0	0	0
3	is	0	0	1	0	0	0	0
4	really	0	0	0	1	0	0	0
5	wonderful	0	0	0	0	1	0	0
6	good	0	0	0	0	0	1	0
7	awful	0	0	0	0	0	0	1

Step - 1

The product is really good The product is wonderful The product is awful

Step - 4

Context Words	Target Word
The,is	Product
product,really	is
Is,good	really
Good,product	the

Step - 5

Training the model

Step 4: Represents Training data

One Hot Encoding

1	The	1	0	0	0	0	0	0
2	product	0	1	0	0	0	0	0
3	is	0	0	1	0	0	0	0
4	really	0	0	0	1	0	0	0
5	wonderful	0	0	0	0	1	0	0
6	good	0	0	0	0	0	1	0
7	awful	0	0	0	0	0	0	1

Training Data

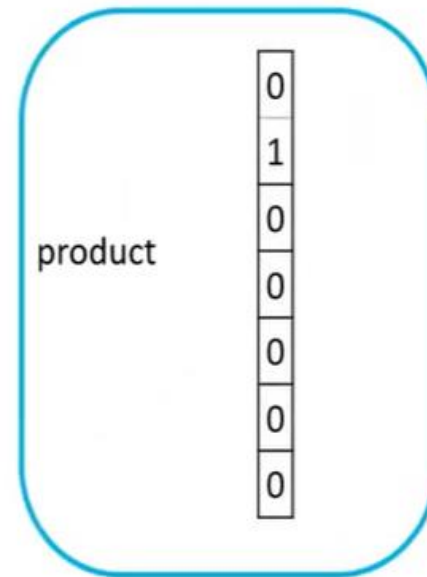
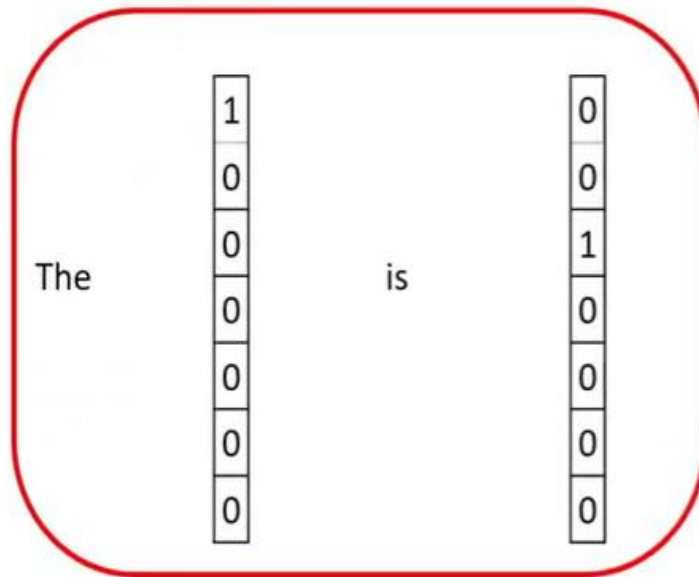
Context Words	Target Word
The,is	product
product,really	is
Is,good	really
Good,product	the

One Hot Encoding

1	The	1	0	0	0	0	0	0
2	product	0	1	0	0	0	0	0
3	is	0	0	1	0	0	0	0
4	really	0	0	0	1	0	0	0
5	wonderful	0	0	0	0	1	0	0
6	good	0	0	0	0	0	1	0
7	awful	0	0	0	0	0	0	1

Training Data

Context Words	Target Word
The,is	Product
product,really	is
Is,good	really
Good,product	the



One hot encoding of - The

1
0
0
0
0
0
0
0

One hot encoding of - is

0
0
1
0
0
0
0
0

Input Layer

14 * 5

5 * 7

Hidden Layer

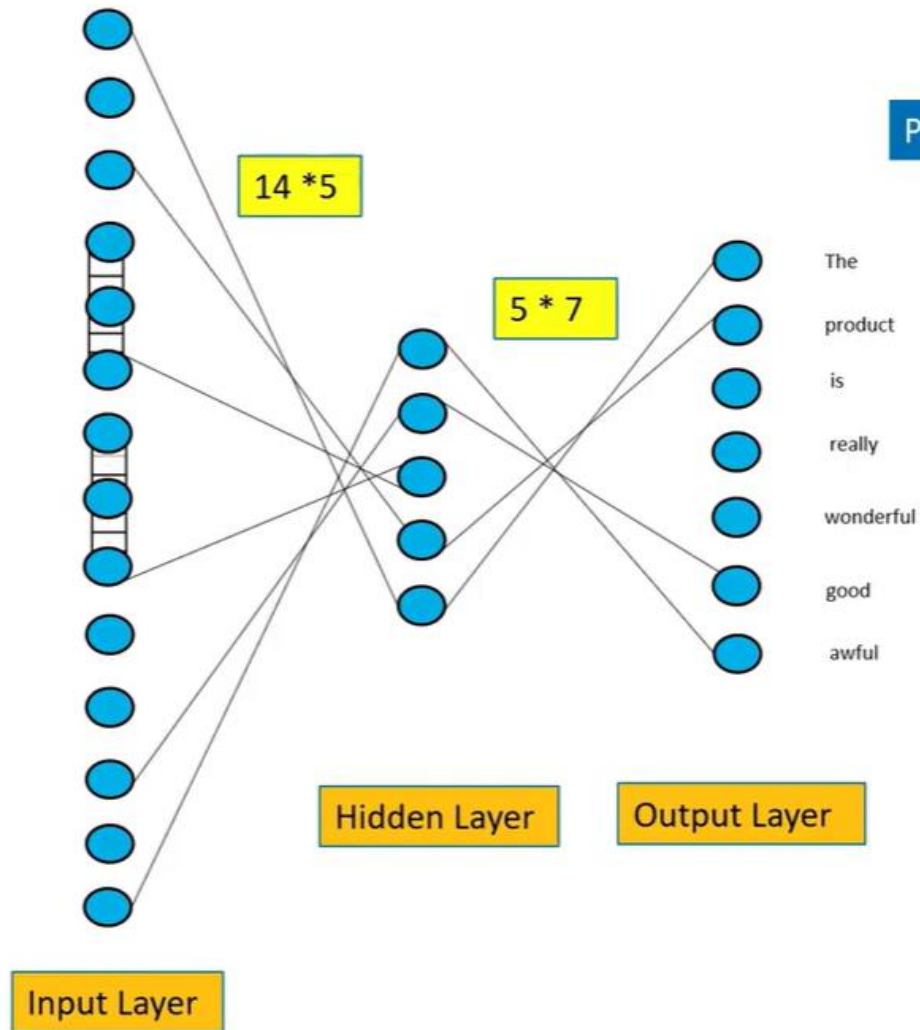
Output Layer

The
product
is
really
wonderful
good
awful

O/P : indicates 7 unique words

One hot encoding of - The

One hot encoding of - is



Predicted

Actual

The

0.1

0

product

0.8

1

is

0.2

0

really

0.1

0

wonderful

0.3

0

good

0

0

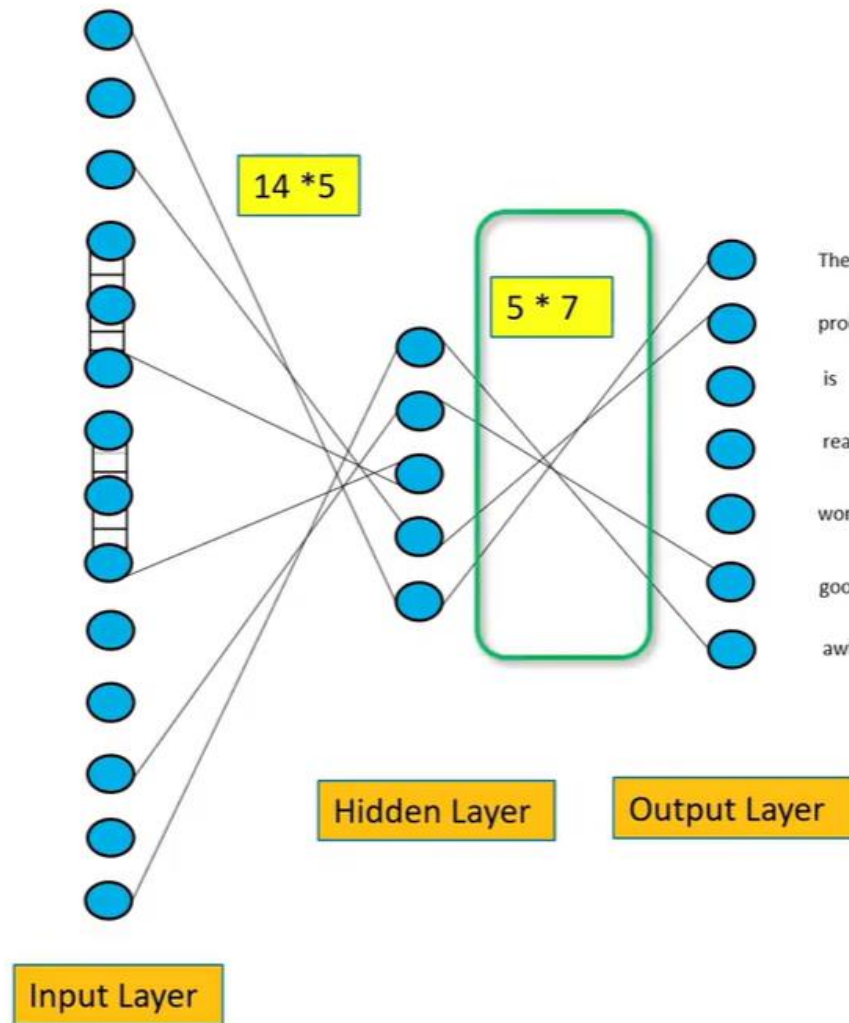
awful

0.1

0

One hot encoding of - The

One hot encoding of - is



Predicted

Actual

The
product
is
really
wonderful
good
awful

0.1
0.8
0.2
0.1
0.3
0
0.1

0
1
0
0
0
0
0

$$5 * 7$$

0.12	0.56	0.23	0.67	0.87	0.9	0.56
0.23	0.45	0.98	0.76	0.98	0.56	0.94
0.45	0.78	0.87	0.62	0.13	0.78	0.1
0.91	0.6	0.24	0.84	0.45	0.67	0.34
0.12	0.87	0.34	0.67	0.78	0.34	0.86

This is the actual Word embedding we are looking for

$7 * 7$

The	1	0	0	0	0	0	0
product	0	1	0	0	0	0	0
is	0	0	1	0	0	0	0
really	0	0	0	1	0	0	0
wonderful	0	0	0	0	1	0	0
good	0	0	0	0	0	1	0
awful	0	0	0	0	0	0	1

X

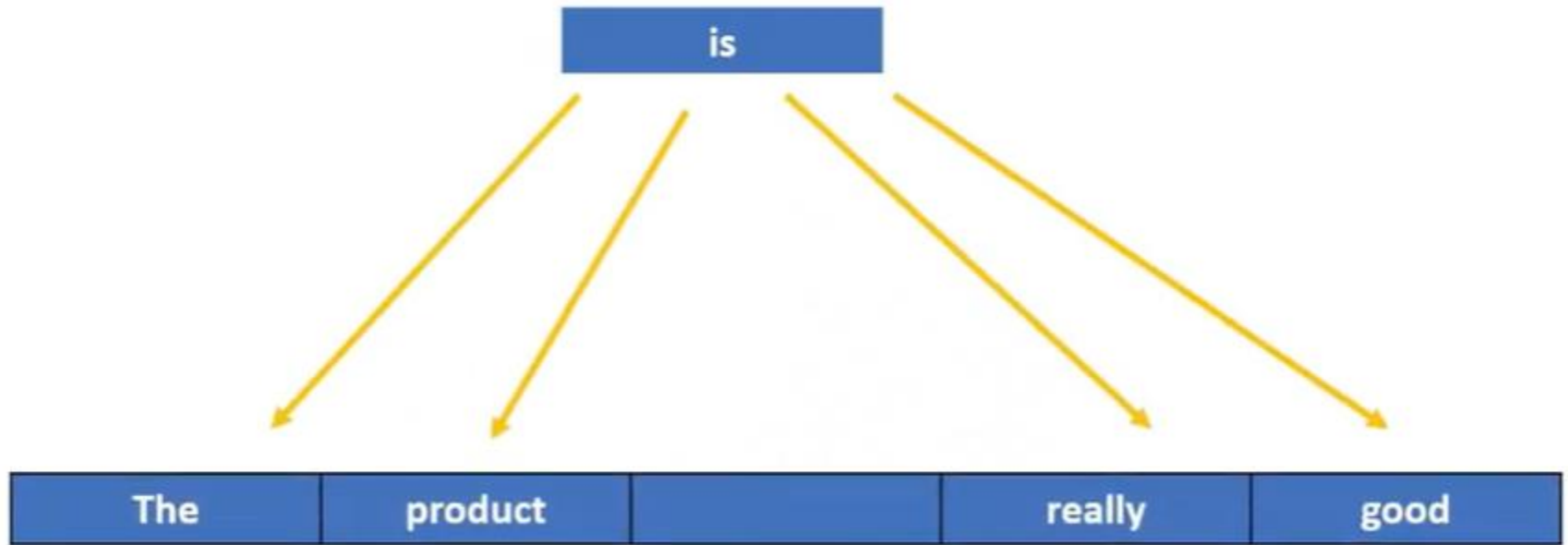
 $(5 * 7)$ Transpose

0.12	0.56	0.23	0.67	0.87	0.9	0.56
0.23	0.45	0.98	0.76	0.98	0.56	0.94
0.45	0.78	0.87	0.62	0.13	0.78	0.1
0.91	0.6	0.24	0.84	0.45	0.67	0.34
0.12	0.87	0.34	0.67	0.78	0.34	0.86

=

 $7 * 5$

SKIP-GRAM



Step - 1

The product is really good The product is wonderful The product is awful

Step - 2

Window Size 1

Step - 3

Get one hot encoding for each word

Step - 1

The product is really good The product is wonderful The product is awful

Step - 3

1	The	1	0	0	0	0	0	0
2	product	0	1	0	0	0	0	0
3	is	0	0	1	0	0	0	0
4	really	0	0	0	1	0	0	0
5	wonderful	0	0	0	0	1	0	0
6	good	0	0	0	0	0	1	0
7	awful	0	0	0	0	0	0	1

Step - 4

Prepare Training data

Step - 1

The product is really good The product is wonderful The product is awful

Step - 4

Input Words	Target Word
product	The
product	is
is	product
is	really

One Hot Encoding

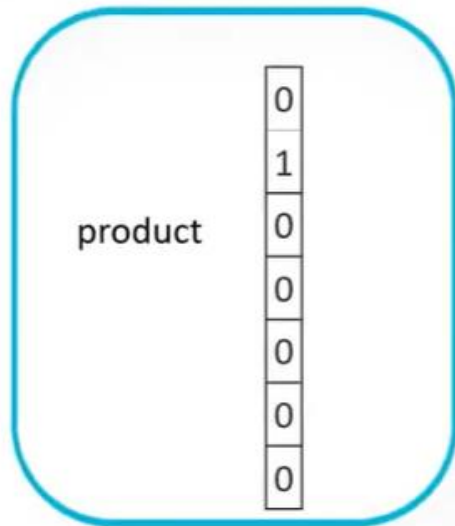
1	The	1	0	0	0	0	0	0
2	product	0	1	0	0	0	0	0
3	is	0	0	1	0	0	0	0
4	really	0	0	0	1	0	0	0
5	wonderful	0	0	0	0	1	0	0
6	good	0	0	0	0	0	1	0
7	awful	0	0	0	0	0	0	1

Training Data

Input Words	Target Word
product	The
product	is
is	product
is	really

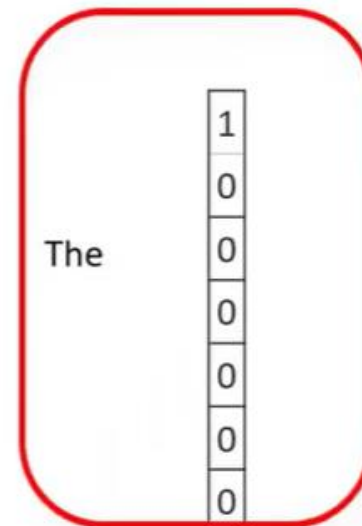
One Hot Encoding

1	The	1	0	0	0	0	0	0
2	product	0	1	0	0	0	0	0
3	is	0	0	1	0	0	0	0
4	really	0	0	0	1	0	0	0
5	wonderful	0	0	0	0	1	0	0
6	good	0	0	0	0	0	1	0
7	awful	0	0	0	0	0	0	1



Training Data

Context Words	Target Word
The, is	Product
product, really	is
Is, good	really
Good, product	the



One hot encoding of - Product

1
0
0
0
0
0
0
0

7 * 5

5 * 7

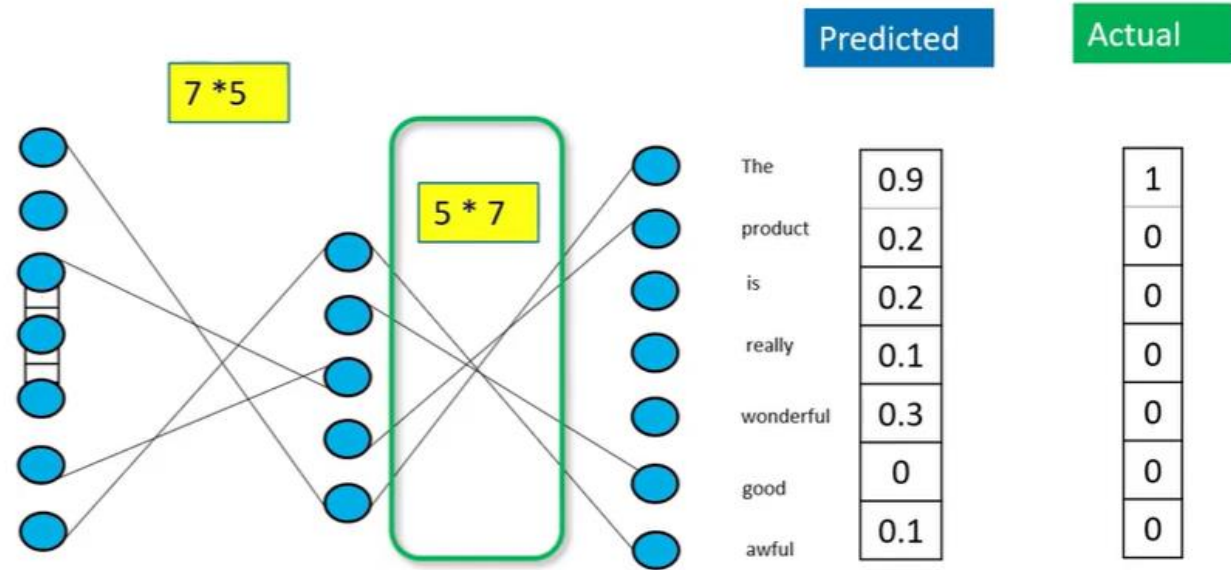
Input Layer

Hidden Layer

Output Layer

The
product
is
really
wonderful
good
awful

One hot encoding of - Product



7 * 7

The	1	0	0	0	0	0	0
product	0	1	0	0	0	0	0
is	0	0	1	0	0	0	0
really	0	0	0	1	0	0	0
wonderful	0	0	0	0	1	0	0
good	0	0	0	0	0	1	0
awful	0	0	0	0	0	0	1



(5 * 7) Transpose

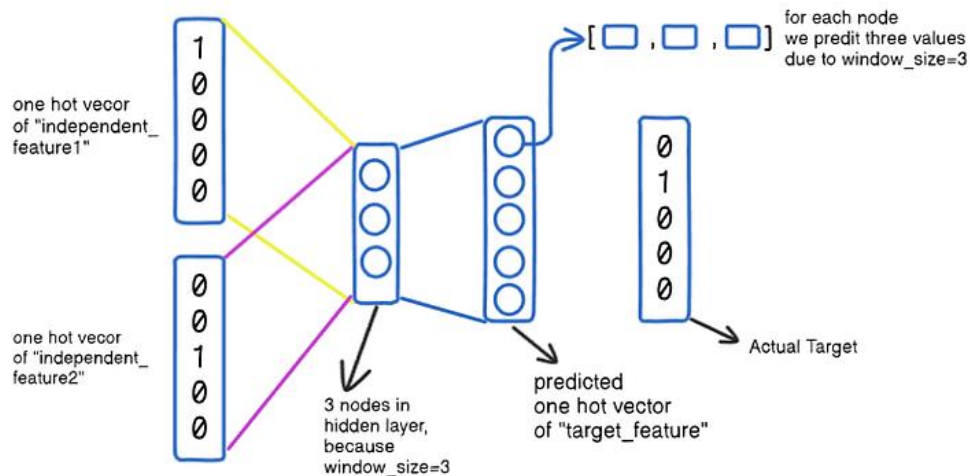
0.12	0.56	0.23	0.67	0.87	0.9	0.56
0.23	0.45	0.98	0.76	0.98	0.56	0.94
0.45	0.78	0.87	0.62	0.13	0.78	0.1
0.91	0.6	0.24	0.84	0.45	0.67	0.34
0.12	0.87	0.34	0.67	0.78	0.34	0.86



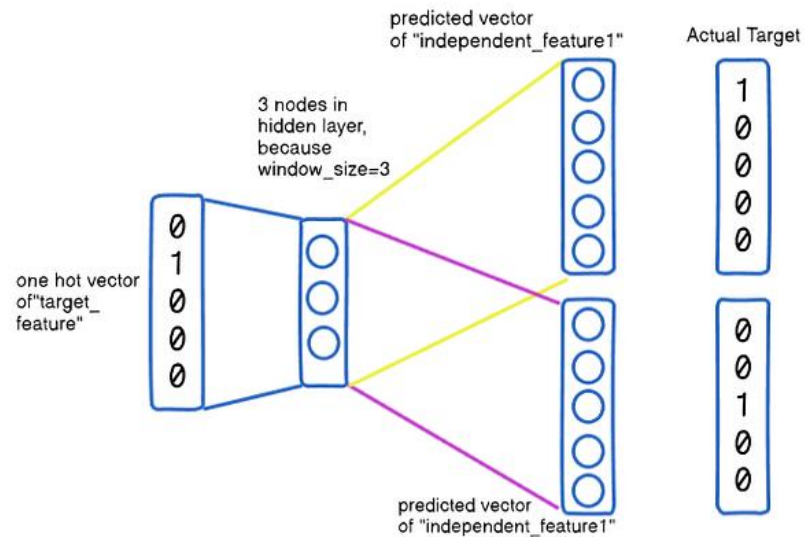
7 * 5

Word2vec

CBOW



Skip-gram



Model	What it predicts
CBOW	Predicts the target (center) word from context words
Skip-Gram	Predicts context words from the target (center) word

Feature	CBOW	Skip-Gram
Prediction Direction	Context $\rightarrow$ Target	Target $\rightarrow$ Context
Training Speed	Faster	Slower
Performance on Small Data	Average	Better
Handles Rare Words	Poor	Very good
Computational Cost	Low	High
Best For	Large datasets	Small datasets / rare words

